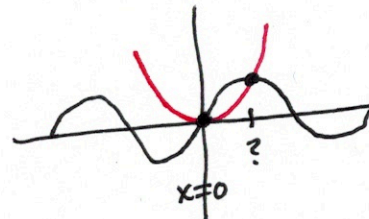


Section 4.8 Newton's Method

Solve $x^2 = \sin x$



$$x^2 - \sin x = 0$$

Let $f(x) = x^2 - \sin x$ (when $f(x) = 0$, $y = 0$)

$$f'(x) = 2x - \cos x$$

Guess $x = 1 \longrightarrow y = 1^2 - \sin(1) = 0.159$

$$f'(1) = 2(1) - \cos(1)$$

$$m = 1.460$$

$$y - 0.159 = 1.460(x - 1)$$

$$y - 0.159 = 1.46x - 1.46$$

$$y = +1.46x - 1.301$$

$$0 = 1.46x - 1.301$$

$$\frac{1.301}{1.46} = x \longrightarrow x = 0.891$$

$$X - \frac{f(x)}{f'(x)}$$

$$X - \frac{x^2 - \sin x}{2x - \cos x}$$

Guess $x_0 = 1$

$$x_1 = .891$$

$$x_2 = .877$$

$$x_3 = .877$$

$$x = .877$$

Ex

$$\sqrt{x} - 1 = \ln x$$

Find one
Solution other
than $x=1$.

$$\sqrt{x} - 1 - \ln x = 0$$

$$f(x) = \sqrt{x} - 1 - \ln x$$

$$f'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{x}$$

$$x = \frac{\sqrt{x} - 1 - \ln x}{\frac{1}{2\sqrt{x}} - \frac{1}{x}}$$

Guess $x_0 = 12.5$

$$x_1 = 12.340$$

$$x_2 = 12.340$$

$$x = 12.340$$